

# Sec 2 Geography Cheat Sheet

Chapters 7-11 and the geographical investigation skills in the EOY scope, with key terms, Singapore examples and limitations. The "How to answer" section explains each question type, and "Practice questions" has 20 exam-style questions with model answers.

## Cities & housing Chapters 7-9 · 8 topics

### What makes a city

CH 7

A **city** is a large urban settlement. Geographers recognise one by these features:

| FEATURE                 | WHAT IT MEANS                                                              |
|-------------------------|----------------------------------------------------------------------------|
| Large population        | hundreds of thousands to millions of people                                |
| High population density | many people per km <sup>2</sup> , so buildings are tall and close together |
| Non-farming jobs        | most people work in industry (secondary) and services (tertiary)           |
| Many functions          | residential, commercial, industrial, recreational, administrative          |
| Good infrastructure     | roads, public transport, water, electricity, sewers, internet              |

- **Urban area:** built-up area such as a town or city. **Rural area:** countryside with farms and villages
- **Population density** = number of people ÷ land area (people per km<sup>2</sup>)
- **Land use:** how land is used, e.g. residential, commercial, industrial, transport, green space

**Watch out:** a big population alone doesn't make a city. Mention density and the kind of jobs too.

### Urbanisation & urban challenges

CH 7

- **Urbanisation:** the increase in the proportion of people living in urban areas
- Caused by **rural-urban migration** and **natural increase** (more births than deaths in cities)

| PUSH FACTORS (RURAL AREAS)    | PULL FACTORS (CITIES)               |
|-------------------------------|-------------------------------------|
| few jobs, low pay             | more jobs, higher pay               |
| poor schools and healthcare   | better schools, hospitals           |
| crop failure, natural hazards | better services, water, electricity |
| little entertainment          | "bright lights", more opportunities |

#### Challenges when cities grow too fast:

| CHALLENGE           | CAUSES → EFFECTS                                                                  |
|---------------------|-----------------------------------------------------------------------------------|
| Housing shortage    | demand exceeds supply → high prices, overcrowding, slums and squatter settlements |
| Traffic congestion  | more vehicles than road space → longer journeys, lost productivity, stress        |
| Pollution           | vehicles, factories, waste → air and water pollution, health problems             |
| Strain on services  | not enough water, sanitation, schools, hospitals                                  |
| Loss of green space | forests cleared for building → less biodiversity, hotter city (urban heat)        |

**Watch out:** "urbanisation" is about the share of people rising. "Urban growth" is the city getting physically larger. Don't mix them up.

### Sustainable & liveable cities

CH 7

**Sustainable development** meets the needs of people today without compromising the ability of future generations to meet their own needs.

| PILLAR   | A SUSTAINABLE CITY...                                             | SINGAPORE EXAMPLE                                         |
|----------|-------------------------------------------------------------------|-----------------------------------------------------------|
| Economic | provides jobs, attracts investment, keeps living costs manageable | business hubs outside the CBD (e.g. Jurong Lake District) |
| Social   | affordable housing, good services, safety, community bonds        | HDB new towns with schools, clinics, community centres    |

|                      |                                                              |                                                     |
|----------------------|--------------------------------------------------------------|-----------------------------------------------------|
| Environmental spaces | limits pollution, saves energy and water, keeps green spaces | parks and park connectors, NEWater, green buildings |
|----------------------|--------------------------------------------------------------|-----------------------------------------------------|

**Singapore's Liveability Framework** (Centre for Liveable Cities) aims for three outcomes: **high quality of life**, **competitive economy** and **sustainable environment**. It gets there through **integrated master planning** (planning land use, housing and transport together, long-term) and **dynamic urban governance** (government working with people and businesses).

#### Other cities to quote:

- **Curitiba, Brazil:** bus rapid transit with dedicated bus lanes, large parks that also absorb floodwater, recycling programmes
- **Copenhagen, Denmark:** extensive cycle lanes, so a large share of people cycle to work or school

**Watch out:** link each strategy to a pillar and say how it helps. "Parks are good for the environment" scores low. "Parks absorb carbon dioxide and lower temperatures, making the city cooler" scores well.

## Types of housing

CH 8

| TYPE             | FEATURES                                                                                             | EXAMPLES                                                                                            |
|------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Public housing   | built or funded by the government, sold or rented at subsidised prices                               | HDB flats in Singapore (close to 8 in 10 residents live in one), public rental housing in Hong Kong |
| Private housing  | built by private developers, sold at market prices, often with facilities like pools and gyms        | condominiums, apartments, landed houses (terrace, semi-detached, bungalow)                          |
| Informal housing | built by residents themselves on land they don't own, often without water, sanitation or electricity | slums and squatter settlements, e.g. favelas in Rio de Janeiro, Dharavi in Mumbai                   |

**Why Singapore builds high-rise:** land is scarce, and building upwards houses more people on less land, leaving land for industry, transport and green spaces.

**Building and Construction Authority (BCA):** sets safety standards for buildings, approves building plans, inspects buildings before people move in, and runs the Green Mark scheme for energy- and water-saving buildings.

**HDB history in one line:** HDB was set up in 1960 to replace crowded kampongs and squatter settlements, and it now houses most Singaporeans.

**Watch out:** "squatter settlement" means the residents don't own the land. Not every slum is a squatter settlement, and not every squatter settlement is a slum.

## Spatial distribution of housing

CH 8

**Spatial distribution** is where something is found and the pattern it forms on a map.

| PATTERN   | WHAT IT LOOKS LIKE                           |
|-----------|----------------------------------------------|
| Clustered | grouped closely together in some areas       |
| Dispersed | spread out with large gaps                   |
| Linear    | along a line, e.g. a road, MRT line or coast |
| Uneven    | more in some areas than others               |

#### Housing in Singapore:

- **HDB new towns** ring the island (e.g. Woodlands in the north, Tampines in the east, Jurong West in the west, Sengkang and Punggol in the north-east)
- Each new town has a **town centre** (mall, MRT station, bus interchange), **neighbourhoods** with their own shops, and **precincts** of blocks around shared spaces
- **Private condominiums** are common near the city centre, along the coast and near MRT stations. **Landed estates** are found in areas such as Bukit Timah, Serangoon Gardens and the east coast
- Density is **highest** near town centres and MRT stations, and **lower** in landed estates

**Watch out:** when describing a map, use compass directions and name places. "Most HDB towns are in the north, east and west, away from the Central Area" beats "they are spread out".

## Factors affecting where housing is built

CH 8

| FACTOR                                    | HOW IT AFFECTS HOUSING                                                                                                                                                                                                                |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Physical:</b> relief (landform)        | flat land is cheaper and easier to build on than steep slopes                                                                                                                                                                         |
| <b>Physical:</b> drainage, flooding       | well-drained land is safer. Low-lying, flood-prone land is avoided or needs drains                                                                                                                                                    |
| <b>Human:</b> land availability and price | scarce, expensive land leads to high-rise housing. Cheaper land further out allows larger towns                                                                                                                                       |
| <b>Human:</b> accessibility               | people want to live near MRT stations, bus routes and expressways                                                                                                                                                                     |
| <b>Human:</b> jobs and amenities          | housing is built near workplaces, schools, shops and hospitals                                                                                                                                                                        |
| <b>Human:</b> zoning and planning         | <b>Zoning</b> assigns each area of land to one use (residential, industrial, commercial) in the Master Plan. Homes go in residential zones, away from polluting industry and near shops and transport. HDB decides where new towns go |

### EXAMPLE: WHY PUNGGOL WAS DEVELOPED

Large area of undeveloped land in the north-east, planned from the start with an LRT and MRT line, a waterway, and parks, to relieve housing demand in older towns.

**Watch out:** split your answer into physical and human factors. Markers often want one of each.

## Housing challenges

CH 9

| CHALLENGE                 | WHAT IT MEANS                                  | EFFECTS                                                     |
|---------------------------|------------------------------------------------|-------------------------------------------------------------|
| Shortage                  | not enough homes for the population            | long waiting times, overcrowding, slums                     |
| Affordability             | homes cost too much compared with income       | young couples can't buy, people live far from work          |
| Poor quality / inadequacy | unsafe buildings, no clean water or sanitation | disease, fire risk, low quality of life                     |
| Ageing estates            | old blocks and facilities wear out             | need upgrading or redevelopment, residents may need to move |
| Environmental impact      | building uses land, energy and materials       | loss of green space, more emissions                         |

**Sustainable housing** is housing that is **affordable**, **adequate** (safe, enough space, with basic services), **accessible** (near transport, jobs, amenities) and **environmentally friendly**, now and in the future.

**Watch out:** "adequate" is not the same as "enough homes". It means each home is good enough to live in.

## Strategies for sustainable housing

CH 9

| STRATEGY                                          | HOW IT HELPS                                                                                                     | LIMITATION                                                         |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>Public housing</b> (HDB flats, Build-To-Order) | large supply of affordable homes. Grants help first-time buyers                                                  | high cost to government. Waiting time of a few years for BTO flats |
| <b>Integrated town planning</b>                   | schools, clinics, shops, parks and transport built with the flats, so homes are accessible                       | takes many years. Early residents may lack facilities at first     |
| <b>Ethnic Integration Policy</b>                  | ethnic quotas per block and neighbourhood prevent enclaves and build social harmony                              | can make it harder for some owners to sell their flats             |
| <b>Upgrading programmes</b>                       | repair and improve older flats and estates (e.g. new lifts, repairs, better facilities) so they last longer      | costly. Disruption during works                                    |
| <b>Green features</b>                             | solar panels on HDB roofs, rainwater collection, greenery. Tengah "forest town" and Punggol "eco-town"           | higher building cost. Benefits take time to show                   |
| <b>Slum upgrading</b> (other cities)              | adds water, sewers, roads and legal land titles without evicting residents, e.g. Favela-Bairro in Rio de Janeiro | needs long-term funding. Can raise rents and push out the poorest  |

**Criteria for evaluating a strategy** (use these in the essay):

- **Cost:** who pays and is it affordable in the long run?
- **Scale:** how many people benefit?
- **Time:** how quickly does it work and how long do benefits last?
- **Pillars:** does it help social, economic and environmental needs?

**Watch out:** always give a limitation. An answer that only praises a strategy can't reach the top level.

## Transport modes & networks

CH 10

| TERM                               | MEANING                                                                                                             |                                                            |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| Transport system                   | the network of routes, vehicles, stops and rules that move people and goods                                         |                                                            |
| Node                               | a point where routes meet or journeys start and end (station, interchange, junction)                                |                                                            |
| Link                               | the route between two nodes (road, rail line)                                                                       |                                                            |
| Accessibility                      | how easily a place can be reached                                                                                   |                                                            |
| Connectivity                       | how well the nodes are linked. More links = better connectivity                                                     |                                                            |
| Intermodal connectivity            | linking different modes (MRT, bus, taxi, bicycle) so people can switch easily, e.g. at integrated transport hubs    |                                                            |
| Point-to-point network             | direct routes between each pair of places. Fast, no transfers, but needs many routes                                |                                                            |
| Hub-and-spoke network              | routes run through a central hub. Fewer routes and efficient, but passengers transfer, e.g. Changi Airport as a hub |                                                            |
| MODE                               | ADVANTAGES                                                                                                          | DISADVANTAGES                                              |
| MRT / LRT (rail)                   | fast, carries many people, no traffic jams, less pollution per person                                               | very expensive to build, fixed routes                      |
| Bus                                | flexible routes, reaches more places, cheaper to set up                                                             | affected by traffic, slower                                |
| Car                                | door-to-door, comfortable, flexible                                                                                 | congestion, pollution, costly, uses road and parking space |
| Walking, cycling (active mobility) | no emissions, healthy, cheap                                                                                        | short distances only, weather (heat, rain)                 |

**Network patterns:** **radial** (routes spread out from a centre, like spokes of a wheel) and **grid** (routes cross at right angles).

## Spatial distribution of transport in Singapore

CH 10

- MRT lines are **densest in the Central Area**, where many lines meet, and spread outwards to new towns in a roughly **radial** pattern
- The **Circle Line** links lines outside the city centre so people don't all have to travel through the centre
- Major interchanges sit where lines cross (e.g. Dhoby Ghaut, Outram Park, Jurong East)
- New towns have **integrated transport hubs** linking MRT, bus interchange and a mall under one roof
- **LRT** loops serve towns like Sengkang, Punggol and Bukit Panjang, feeding residents to the MRT
- Expressways link the regions to the Central Area and to each other

### Why transport is distributed this way:

- **Population:** routes serve where many people live (HDB new towns)
- **Economic activity:** routes lead to where people work (CBD, industrial estates, business parks)
- **Planning:** new towns are planned with MRT and bus links from the start
- **Physical features:** routes avoid or tunnel under nature reserves and water catchments

**Watch out:** give map evidence: name lines or stations, count them in an area, and use compass directions.

## Transport problems

CH 11

| PROBLEM            | CAUSES                                                    | EFFECTS                                                      |
|--------------------|-----------------------------------------------------------|--------------------------------------------------------------|
| Traffic congestion | too many vehicles, peak-hour travel, roadworks, accidents | time and fuel wasted, lower productivity, stress             |
| Air pollution      | vehicle exhaust                                           | breathing problems, more greenhouse gases and climate change |
| Noise pollution    | heavy traffic, trains, aircraft                           | stress, poor sleep                                           |
| Accidents          | speeding, crowded roads                                   | injuries, deaths, more congestion                            |
| Land use           | roads and car parks take up space                         | less land for homes, parks and industry                      |

Overcrowding on public transport

demand at peak hours exceeds capacity

uncomfortable journeys, more people choose to drive

**Watch out:** an "explain" answer needs the chain: cause → problem → effect on people or the environment.

## Managing transport sustainably

CH 11

Two approaches: reduce the **demand** for car travel, and increase the **supply** of better options.

| STRATEGY                                                                                    | HOW IT WORKS                                                                                      | LIMITATION                                                              |
|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| <b>Vehicle Quota System and COE</b><br>(since 1990)                                         | limits the number of new vehicles. Buyers bid for a Certificate of Entitlement, valid 10 years    | makes cars very expensive, unfair to lower-income families              |
| <b>Electronic Road Pricing (ERP)</b> (since 1998, replacing the 1975 Area Licensing Scheme) | charges drivers on busy roads at peak times, so some change route, time or mode                   | drivers who can afford it still drive. Traffic can shift to other roads |
| <b>Improving public transport</b>                                                           | new MRT lines, more buses, integrated hubs, concession fares make it a better choice than driving | very expensive, takes years to build, crowding at peak hours            |
| <b>Active mobility</b>                                                                      | cycling paths, park connectors, covered walkways encourage walking and cycling                    | hot, rainy weather. Only suits short trips                              |
| <b>Land use planning</b>                                                                    | homes, jobs and services close together (e.g. regional centres) so people travel less             | long-term, takes decades                                                |
| <b>Cleaner vehicles and technology</b>                                                      | electric buses and cars, EV chargers, real-time traffic and bus arrival apps                      | high cost of new vehicles and chargers                                  |

**Singapore's Land Transport Master Plan 2040:** "20-minute towns" and a "45-minute city", meaning most peak-hour journeys by walking, cycling and public transport should take under 20 minutes to the nearest town centre and under 45 minutes to anywhere in the city. Motto: *Walk Cycle Ride*.

**Examples elsewhere:** **London** congestion charge (since 2003) charges cars entering central London. **Curitiba** bus rapid transit uses dedicated lanes and tube stations for fast boarding.

**Watch out:** ERP charges you each time you pass a gantry at certain times. COE is a certificate you must buy before you can own a car. Don't mix them up.

## Geographical investigation steps

GI

1. **Guiding question or hypothesis:** what you want to find out, e.g. "People living further from an MRT station are less likely to take public transport"
2. **Data collection:** **primary** data you collect yourself (questionnaire, interview, observation, traffic count, field sketch, photographs) and **secondary** data from others (maps, census, websites)
3. **Sampling:** choose who or where to collect data from
4. **Data representation:** graphs, tables, maps
5. **Analysis:** describe patterns and relationships, explain them
6. **Conclusion:** answer the guiding question or accept/reject the hypothesis
7. **Evaluation:** limitations and how to improve

| METHOD                     | GOOD FOR                          | LIMITATION                               |
|----------------------------|-----------------------------------|------------------------------------------|
| Questionnaire              | many people, quick, easy to count | people may not answer honestly or at all |
| Interview                  | detailed opinions                 | takes long, small sample                 |
| Traffic / pedestrian count | numbers at a place and time       | varies with day, time and weather        |

## Sampling & reliability

GI

| METHOD      | HOW                                                                      | STRENGTH               | WEAKNESS                              |
|-------------|--------------------------------------------------------------------------|------------------------|---------------------------------------|
| Random      | everyone has an equal chance, e.g. random number generator               | no bias                | may miss some groups                  |
| Systematic  | at regular intervals, e.g. every 5th person or every 50 m                | quick, spread evenly   | a hidden pattern can skew results     |
| Stratified  | split into groups (age, gender, housing type), sample each in proportion | represents every group | needs information on the groups first |
| Convenience | whoever is easiest to reach                                              | fast, easy             | biased, not representative            |

- **Reliability:** similar results if the investigation is repeated
- **Accuracy:** how close the results are to the true value

**To improve reliability:** use a larger sample, repeat readings and take an average, collect at the same time and in similar conditions, use the same questions for everyone, do a pilot test first.

**Watch out:** give the reason. "Use a larger sample so that unusual answers have less effect on the results."

## Data representation & scatter graphs

GI

| DATA                               | BEST WAY TO SHOW IT                               |
|------------------------------------|---------------------------------------------------|
| Parts of a whole (%)               | pie chart, divided bar graph                      |
| Comparing categories               | bar graph                                         |
| Change over time                   | line graph                                        |
| Relationship between two variables | scatter graph                                     |
| Values across places               | choropleth map (shading), proportional symbol map |

**Drawing a scatter graph:** independent variable on the x-axis (e.g. distance), dependent on the y-axis (e.g. number of users). Title, labelled axes with units, even scales. Plot with small crosses. Draw one straight **best-fit line** through the middle of the points.

- **Positive relationship:** as x increases, y increases
- **Negative relationship:** as x increases, y decreases
- **No relationship:** no clear pattern
- **Strong** if points lie close to the line, **weak** if widely spread
- **Anomaly:** a point far from the line. Suggest a reason for it

**Watch out:** a relationship doesn't prove one thing causes the other. Say "there is a relationship", not "distance causes".

## Command words & marks

EXAM

| COMMAND WORD                           | WHAT TO DO                                                              | MARKS GUIDE                       |
|----------------------------------------|-------------------------------------------------------------------------|-----------------------------------|
| <b>Identify / State / Name</b>         | a short answer, no explanation                                          | 1 mark per point                  |
| <b>Describe</b>                        | what it looks like: pattern, trend, figures with units                  | 1 mark per point or data quote    |
| <b>Compare</b>                         | similarities and differences, linked in one sentence with data for both | 1-2 marks per comparison          |
| <b>Explain</b>                         | give reasons: point → why/how → example                                 | about 2 marks per developed point |
| <b>Suggest</b>                         | a sensible reason or idea, even if not in the textbook                  | 1-2 marks per suggestion          |
| <b>Evaluate / How far do you agree</b> | both sides, then a judgement with reasons                               | marked by level                   |

**Before writing:** circle the command word, underline the topic, note the marks. Marks tell you how many points to make.

**Time:** about 1 minute per mark, plus time to plan the essay.

**Watch out:** "Describe" doesn't need reasons, and "Explain" doesn't need a description. Writing the wrong one earns nothing.

## Describe & compare data

EXAM

Use **TEA**: **T**rend (overall pattern), **E**vidence (quote data with units), **A**nomaly (anything that doesn't fit).

### QUESTION (3 MARKS)

Describe the change in the number of cars in City X from 2010 to 2025 shown in Fig. 1.

### MODEL ANSWER

The number of cars increased overall [T]. It rose from 400 000 in 2010 to 650 000 in 2025 [E]. However, it fell slightly between 2018 and 2020, from 560 000 to 540 000 [A].

**Compare:** link both items in one sentence using *whereas*, *while*, *both*, *higher than*, *twice as much*. Give data for both and the difference.

### QUESTION (2 MARKS)

Compare the use of public transport in Town A and Town B.

### MODEL ANSWER

Town A has a higher share of residents using public transport (70%) whereas Town B has only 45%, a difference of 25 percentage points. Both towns have the bus as their most common mode.

**Watch out:** copy the units from the graph (% , km, thousands). Numbers without units can lose marks. Two separate descriptions are not a comparison.

## Explain questions

EXAM

Use **PEE** for each reason: **P**oint → **E**xplain how or why → **E**xample (a Singapore case or data from the question).

### QUESTION (4 MARKS)

Explain two reasons why most housing in Singapore is high-rise.

### MODEL ANSWER

**P:** Singapore has very limited land. **E:** Building upwards houses many families on a small plot of land, leaving space for industry, roads and parks. **E:** An HDB block of 30 storeys can house hundreds of households on one site.

**P:** There is high demand for housing from a large population. **E:** High-rise flats let the government provide enough homes quickly and at lower cost per home, keeping them affordable. **E:** Close to 8 in 10 residents live in HDB flats.

### QUESTION (4 MARKS)

Explain how traffic congestion affects people living in a city.



### MODEL ANSWER

Congestion makes journeys longer, so people spend more time travelling and less time working or with family, lowering productivity. Idling vehicles release more exhaust fumes, which pollutes the air and can cause breathing problems for people living near busy roads.

**Watch out:** one developed reason beats three one-liners. If the question says "two reasons", write two paragraphs, not four bullet points.

## Maps & photographs

EXAM

### Describing a distribution on a map:

1. Name the pattern: clustered, dispersed, linear, uneven
2. Say where: use compass directions and place names
3. Give evidence: count features or name examples
4. Point out any exception

### MODEL ANSWER

MRT stations are clustered in the Central Area, where 12 of the 20 stations shown are found. Outside the centre they are distributed linearly along the lines to the north, east and west. There are few stations in the central catchment area.

**Describing a photograph:** work from **foreground** (front) to **background** (back). Describe what you can see (building height, density, greenery, transport, people), not what you guess.

### MODEL ANSWER

In the foreground there is a wide road with heavy traffic and a bus lane. In the background there are closely packed high-rise blocks of about 20–30 storeys with little green space between them.

**Watch out:** only describe what is visible. Use "suggest" words (may, likely) if the question asks you to infer.

## Investigation skills questions

EXAM

### QUESTION (2 MARKS)

Suggest a suitable sampling method for a survey of how residents in a new town travel to work. Justify your choice.

### MODEL ANSWER

Stratified sampling, by dividing residents by housing type (e.g. HDB and private) and surveying each group in proportion. This ensures all groups are represented, since car ownership may differ between them.

### QUESTION (2 MARKS)

Explain how the students could make their traffic count more reliable.

### MODEL ANSWER

They could repeat the count at the same time on three different weekdays and take the average. This reduces the effect of an unusual day, such as one with an accident or heavy rain.

### QUESTION (3 MARKS)

Describe the relationship shown in the scatter graph.

### MODEL ANSWER

There is a negative relationship: as distance from the MRT station increases, the percentage of residents using public transport decreases. At 200 m it is 80%, but at 1.5 km it is only 30%. Point K is an anomaly, as it has 75% at 1.2 km, possibly because a feeder bus stops there.

**Limitations to mention:** small sample, only one day or time, people answering dishonestly, weather, only one location.

## Housing evaluation essay (LDQ)

EXAM

The EOY essay is a level-descriptor question on **Chapter 9 housing**. Plan for 3–4 minutes before writing.

1. **Introduction (2–3 lines):** define sustainable housing and state your stand
2. **Paragraph 1:** the strategy in the question. Point → how it makes housing sustainable → Singapore example → one limitation
3. **Paragraph 2:** another strategy, same structure
4. **Conclusion:** weigh them with a criterion (cost, scale, time, pillars) and give a clear judgement

### LEVEL WHAT THE ANSWER DOES

Level 1 describes one or two strategies with little explanation or example

|         |                                                                                                     |
|---------|-----------------------------------------------------------------------------------------------------|
| Level 2 | explains strategies on both sides with examples, but judgement is weak or missing                   |
| Level 3 | explains both sides with examples and limitations, and reaches a supported judgement using criteria |

### SAMPLE QUESTION

"Public housing is the most effective strategy to achieve sustainable housing in Singapore." How far do you agree?  
Explain your answer.

### MODEL ANSWER

**Introduction.** Sustainable housing is housing that is affordable, adequate, accessible and environmentally friendly for people now and in the future. I agree to a large extent that public housing is the most effective strategy, although green features are also needed.

**Paragraph 1 (for).** Public housing makes homes affordable for most people. HDB builds flats on a large scale and sells them at subsidised prices, and first-time buyers receive housing grants. As a result, close to 8 in 10 residents live in HDB flats, so very few Singaporeans are without a home. New towns are also planned with schools, clinics and MRT stations, making homes accessible. However, public housing costs the government a great deal, and buyers may wait a few years for a Build-To-Order flat.

**Paragraph 2 (other strategy).** Green features also make housing sustainable by reducing its impact on the environment. For example, solar panels on HDB rooftops generate clean electricity, and Tengah is designed as a "forest town" with green corridors that keep it cooler. However, these features raise building costs and mainly help the environmental pillar, not affordability.

**Conclusion.** Public housing is the most effective strategy because it benefits the largest number of people and meets the social and economic needs of affordable, adequate and accessible homes. Green features are important, but they work best as part of public housing projects rather than on their own. Therefore I agree to a large extent.

**Watch out:** a judgement without a reason stays at Level 2. End with "because..." and name the criterion, e.g. "it benefits the largest number of people".

## Exam-day checklist

EXAM

- Read every question first. Start with the ones you know best
- Circle the command word, underline key words and note the marks
- Match points to marks: a 4-mark "explain" needs two developed reasons
- Quote data with units from every figure you are given
- Use geography terms: accessibility, density, sustainable, spatial distribution, congestion
- Give a Singapore example wherever you can: HDB, Tengah, Punggol, ERP, COE, MRT, LTMP 2040
- Save time for the essay. Plan it before writing
- Leave 5 minutes to check you've answered every part of every question

Practice: cities & urbanisation

CH 7

**3m** Fig. 1 shows the percentage of China's population living in urban areas from 1980 to 2020. Describe the trend.

The percentage of people living in urban areas in China increased steadily over the period [trend]. It rose from about 19% in 1980 to about 64% in 2020, an increase of about 45 percentage points [evidence]. The increase was fastest after 2000 [detail from graph].

**Marking tip:** one mark for the trend, one for quoting start and end values with %, one for a further detail such as the fastest period. Always use the actual figures from the graph given.

**4m** Explain how the expansion of cities harms forests and water resources.

**Forests:** when cities expand, forests are cleared to make land for housing, roads and industries, and trees are cut for timber. This destroys the habitats of plants and animals, reducing biodiversity, and fewer trees are left to absorb carbon dioxide.

**Water:** rivers are polluted by soil washed off construction sites and by sewage and waste from homes and factories, making the water unsafe to use. Land is also covered with concrete, so less rainwater soaks into the ground to refill groundwater and more runs off the surface, increasing the risk of floods.

**Marking tip:** 2 marks for forests and 2 for water. Writing about only one caps the answer at 2 marks.

**3m** Explain how extracting natural resources for housing developments can harm the environment.

Building homes needs large amounts of sand, granite and timber. Dredging sand from rivers and coasts disturbs the riverbed and seabed, destroying habitats and causing erosion. Quarrying for granite removes vegetation and creates dust and noise pollution. Logging for timber clears forests, so soil is exposed and washed into rivers, making them muddy.

**Marking tip:** name the resource, how it is extracted, and the environmental effect. Three linked points for 3 marks.

Practice: housing

CH 8–9

**4m** Fig. 2 shows the average price of HDB resale flats from 1990 to 2025. Describe the changes in price.

Overall, the price of HDB resale flats increased over the period [trend]. *Quote the first and last values from the graph, with units, e.g. "from \$X in 1990 to \$Y in 2025" [evidence]. Describe any period where prices fell or stayed the same, with years and values [anomaly]. Say when prices rose fastest [detail].*

**Marking tip:** use TEA: trend, evidence with \$ and years, anomaly. "Changes" means you must describe any fall or flat period, not just the overall rise.

**3m** Photograph A shows a squatter settlement. Describe its characteristics.

The houses are built from makeshift materials such as zinc sheets, wood and plastic. They are packed closely together with no proper roads or open spaces between them. There are no visible piped water, drains or proper sanitation, and the settlement is built on unsafe land such as a riverbank or slope.

**Marking tip:** only describe what can be seen in the photograph. One mark per characteristic.

**3m** Explain how zoning affects the location of housing.

**Zoning** is when the government assigns each area of land to a specific use, such as residential, industrial or commercial, in a land-use plan like Singapore's Master Plan. Housing can only be built in residential zones, which are placed away from polluting industrial zones so residents live in a safe and healthy environment. Residential zones are also placed near commercial areas and transport, so residents can reach shops, jobs and services easily.

**Marking tip:** define zoning, then explain two effects on where homes are built.

**3m Explain how the Building and Construction Authority (BCA) ensures that housing in Singapore is safe and provides a good quality of life.**

The BCA sets building standards, such as for structural and fire safety, that all buildings must meet. It checks and approves building plans before construction starts and inspects buildings before people are allowed to move in, so homes are safe to live in. It also runs the Green Mark scheme, which rewards buildings that save energy and water, making homes more comfortable and environmentally friendly.

**Marking tip:** one mark for each role linked to safety or quality of life.

**4m Photograph B shows a housing estate in Singapore. Identify one environmentally friendly feature and explain how it benefits the environment.**

One feature is **solar panels** on the roofs of HDB blocks. They convert sunlight into electricity to power lifts, corridor lights and pumps. This reduces the need for electricity from burning fossil fuels, so fewer greenhouse gases are released, which helps reduce climate change.

*Other features you might see:* rooftop gardens or vertical greenery (cool the building and reduce air-conditioning use), rainwater collection systems (reduce water use), recycling chutes (reduce waste sent to incineration).

**Marking tip:** 1 mark for identifying the feature, the rest for explaining the benefit step by step.

**2m Explain why it is important to make public housing inclusive.**

Inclusive housing lets people of different races, incomes, ages and abilities live in the same estate. This helps them interact and build understanding, which strengthens social cohesion and prevents groups from being segregated. For example, the Ethnic Integration Policy sets ethnic quotas in each HDB block, and barrier-free ramps and lifts let elderly residents and wheelchair users move around independently.

**Marking tip:** 1 mark for the reason, 1 for an example or elaboration.

## Practice: transport

CH 10-11

**4m Identify and explain two reasons why cities need transport systems.**

**To move people to work and school:** transport systems allow large numbers of people to travel from their homes to workplaces and schools. This keeps businesses running and supports the city's economy.

**To move goods:** transport systems carry food, materials and products into and around the city. Without them, shops and factories could not get supplies, and residents could not get the goods they need.

**Marking tip:** 2 marks per reason: 1 for identifying it, 1 for explaining it.

**3m Table 1 shows the number of vehicles and the greenhouse gas emissions in a city. Describe the relationship.**

There is a positive relationship: as the number of vehicles increases, greenhouse gas emissions increase. *Quote two pairs of values from the table, e.g. "when there were X vehicles, emissions were Y tonnes, but when vehicles rose to X<sub>2</sub>, emissions rose to Y<sub>2</sub>". Point out any year that doesn't fit the pattern.*

**Marking tip:** state the relationship, give two data pairs with units, and name any anomaly.

**3m Explain what intermodal connectivity is and its advantages.**

**Intermodal connectivity** means linking different modes of transport, such as MRT, bus, taxi and bicycle, so people can switch easily between them, for example at an integrated transport hub. This shortens waiting and walking time when transferring, so journeys are faster and more convenient. As a result, more people choose public transport over driving, reducing congestion and pollution.

**Marking tip:** 1 mark for the definition, 2 marks for explained advantages.

**3m Photograph C shows a road in a city. Identify the problem and state two indicators of it.**

The problem is **traffic congestion**. Indicators: long queues of vehicles bumper to bumper, and vehicles filling every lane and moving slowly or not at all.

**Marking tip:** 1 mark for the problem, 1 mark for each indicator you can see in the photo.

**3m** Photograph D shows a measure used on Singapore's roads (an ERP gantry). Identify the measure and explain why it is necessary.

The measure is **Electronic Road Pricing (ERP)**. Drivers are charged each time they pass under a gantry on busy roads during peak hours. This encourages some drivers to travel at other times, use other routes or take public transport, which reduces traffic congestion on the busiest roads.

**Marking tip:** 1 mark for identifying the measure, 2 for explaining how it works and why it is needed.

**2m** Compare a point-to-point network and a hub-and-spoke network. Give an example.

In a **point-to-point** network, routes go directly between each pair of places, so journeys are faster with no transfers, but many more routes are needed. In a **hub-and-spoke** network, routes go through a central hub, so fewer routes are needed and it is more efficient to run, but passengers often have to transfer. For example, Changi Airport works as a hub, with flights from many cities connecting through Singapore.

**Marking tip:** compare in linked sentences (whereas, but), then give an example.

**4m** (i) Explain one strategy cities use to manage transport sustainably. (ii) Evaluate how effective it is.

**(i)** Singapore improves public transport by building more MRT lines and integrated transport hubs. Frequent, well-connected trains and buses give people a fast alternative to driving, so fewer cars are on the road, reducing congestion and emissions.

**(ii)** This is effective because it benefits a large number of people every day and lasts for many years. However, it is very expensive and takes years to build, and trains can still be crowded at peak hours, so some people keep driving.

**Marking tip:** the evaluation needs a strength and a limitation, then a short judgement.

## Practice: transport essay (6m)

CH 11

**6m** "Research and development is the most effective strategy to sustainably manage transport systems." To what extent do you agree? Use examples to support your answer.

**For:** Research and development (R&D) creates new technology that makes transport cleaner and more efficient. For example, Singapore is replacing diesel buses with electric buses, which produce no exhaust fumes on the road, reducing air pollution. Smart traffic lights and real-time bus arrival apps help traffic flow and shorten waiting times. However, new technology is expensive and can take many years to test before it is widely used.

**Against:** Improving public transport may be more effective. Building new MRT lines and integrated transport hubs gives millions of people a fast alternative to driving, so fewer cars are on the road. Its benefits are felt now and by many people. However, it is very costly and takes years to build.

**Judgement:** I agree only to a small extent. Public transport is more effective because it reduces congestion and emissions for the largest number of people today, while R&D is more useful in the long term to make that public transport cleaner.

**Marking tip:** Level 3 needs both sides explained with examples, a limitation for each, and a judgement with a reason ("because it benefits the most people now").

## Practice: investigation skills

GI

**2m** State the stages of a geographical investigation.

(1) Formulate a guiding question or hypothesis. (2) Collect data. (3) Organise and represent the data. (4) Analyse the data and draw conclusions. (5) Reflect on and evaluate the investigation.

**Marking tip:** use the exact stage names from your class notes, as schools word them slightly differently.

**1m** Write a hypothesis for an investigation into how residents in a new town travel to work.

"Residents who live further from an MRT station are less likely to take public transport to work."

**Marking tip:** a hypothesis is a statement you can test, linking two things. It is not a question.

**2m** Suggest two ways to improve the questionnaire used in the investigation.

Use closed questions with answer options (e.g. MRT / bus / car / others), so the answers are quick to give and easy to count. Avoid leading questions such as "Don't you agree that the MRT is better?", which push people towards an answer and make the results biased.

**Marking tip:** each suggestion needs a reason ("so that...").